Data Centric AI

INTRODUCTION

Business Intelligence Group

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DCAI Modules for ISI and DTM Students (enrolled AA 26-27)

DTM

Machine Learning Module I – 30h – (Prof. Francia)

1. Introduction to ML
2. Supervised classification
3. Clustering
4. Association rules

ISI

Machine Learning Course – (Prof. Maltoni)

Machine Learning Module II ≅ Data Centric AI Module I – 30h – (Prof. Golfarelli)

1. Data Understanding
2. Feature selection
3. Feature engineering
4. Outlier detection
5. Bias and fairness
6. Analytics (collaborative filtering, churn prediction, …)
7. Python labs

Data Centric AI Module II – 30h – (Prof. Francia)

1. ML Pipelines
2. More Algorithms and Python Labs
3. ML Explainability
4. Auto ML Techniques

Data Mining Modules for ISI (enrolled before AA 26-27)

Machine Learning Course – (Prof. Maltoni)

Focus on Data handling and management

Data Mining Module I ≅ Data Centric AI Module I – 30h – (Prof. Golfarelli)

1. Data Understanding
2. Feature selection
3. Feature engineering
4. Outlier detection
5. Bias and fairness
6. Analytics (collaborative filtering, churn prediction, …)
7. Python labs

Text Mining – 30h – (Prof. Moro)

1. Text Representation, Retrieval, Classification, Sentiment Analysis and Opinion Mining, Word embeddings
2. Language Models, Transformers, Efficient Attention Mechanisms
3. Generative Large Language Model, Prompt Engineering, Evaluation Methods

Module Goals

Students learn how to improve the effectiveness of AI solutions by focusing on data quality and management.

More specifically, by the end of the course, students will be able to:

- Critically analyze a dataset and ~~enhance~~ migliorare the explainability of AI systems
- Apply techniques for feature selection and engineering
- Perform data curation, cleaning, and augmentation
- Identify and mitigate issues related to data bias
- Apply Auto-ML techniques, and design applications following the ML-Ops approach

Why a system provide some result?

Most of the time, in real projects, you do Data preparation, feature engineering, data cleaning etc etc

Teaching methods and material

Teaching methods: Classroom and laboratory lessons on Python (YOU must focus on Understanding, not on Programming)

All course topics are covered by downloadable slides and Colab notebooks from the teacher's site

DCAI & ML&DM Assessment Method

Exam consists in an oral exam on all the subjects of the course and a project assignment

- The exam dates can be defined with the teachers along the whole academic year

The examiner will be one of the two module's teachers (Prof. Francia or Prof. Golfarelli)

The choice of the assignment must be agreed with the teachers and can be of two types

- Study of an algorithm (not studied along the classrooms) among those in the literature
- Analysis of a data set with following the Crisp methodology

Data Mining Assessment Method

Exam consists in 2 oral exams, one for each module (Prof. Golfarelli and Prof. Moro), on all the subjects of the course and a project assignment

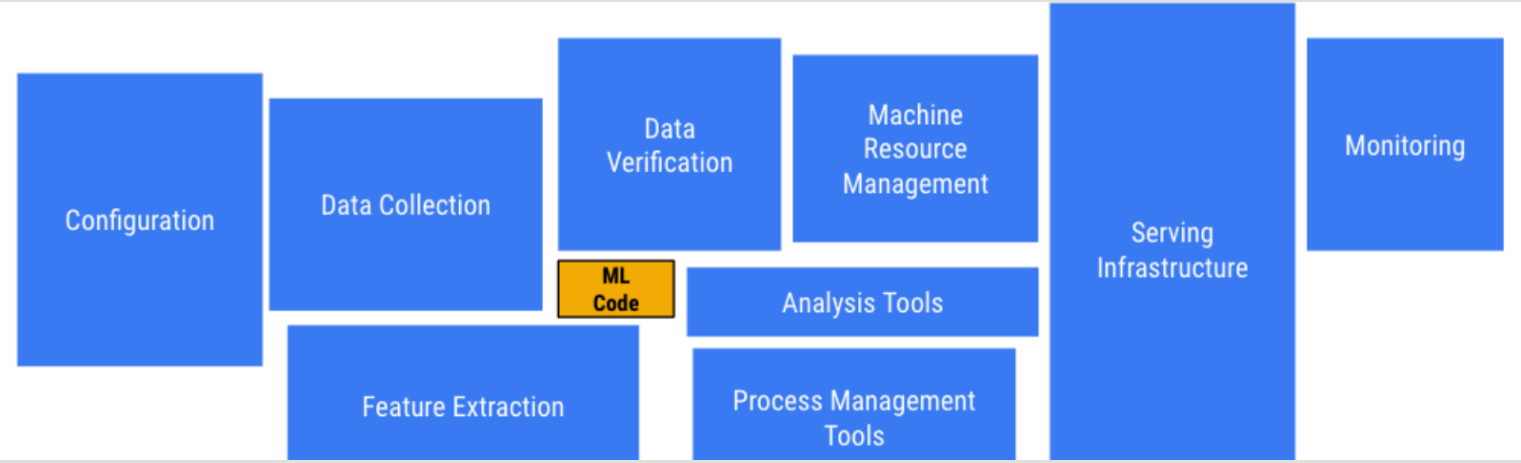
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The Course Program in Details

- Data Understanding and Profiling
- Data Cleaning and Validation
- Feature Engineering
- Dataset Optimization
- Data Quality and Fairness Assessment
- *Graph data and ML models*
- *Privacy-Aware Techniques*
- *Explainability*
- *AutoML and MLOps Applications*
- *Practical Applications*



The Data Centric AI Concept



The concept of Data Centric AI (DCAI) has been coined by Andrew Ng in 2020 (Stanford, ex Google Brain, Baidu, Coursera)

DCAI is an emerging paradigm that emphasizes the importance of data quality and dynamism in AI systems, using an iterative, systematic approach.

In contrast with the Model Centric AI, that merely focusses on model and hyper parameter tuning, DCAI put data at the core of optimization:

It is much easier to achieve better results by improving the data than by improving the model

Paying attention to both data and models is certainly not a new idea, but DCAI places this principle at the heart of ML processes.

The Data Centric AI Principles

Data scientist put a lot of attention not only on model optimization, but also on data optimization (preparation and organisation of data) to improve the perforance

While *model fit* measures how well a model explains observed data, ***data fit*** measures how well the data supports the learning and deployment objectives. Fitting involves several dimensions (Coverage, Representativeness, Bias, Imbalance)



The Data Centric AI Principles



2

Systematic improvement of data consistency denotes a structured approach to identifying, quantifying, and correcting **intra-dataset inconsistencies**, including label ambiguity, annotation drift, conflicting samples, and schema misalignment. This process aims to enforce a stable and task-aligned data interpretation across the entire dataset

The Data Centric AI Principles

Improve and understand the data, then improve and understand the algorithm, then apply and verify then you do it again until you reach a stable point (iterative)



CRISP methodology

3

Miglioramento reciproco

Mutual improvement of model and data through iteration refers to an iterative feedback loop in which model behavior is used to diagnose data deficiencies, and targeted data improvements are then used to refine model performance. Rather than optimizing the model or the dataset in isolation, both evolve jointly through successive cycles of analysis and intervention.

The Data Centric AI Principles



During the iterative approach, we need to put the Human in the loop to get more knowledge about the domain (we can't have high performance without the technical and domain expertise of the human) 4

Human activities such as data collection, annotation, cleaning, and validation are central to system performance. A human-centered approach ~~acknowledges~~ valorizza human expertise and responsibility, ensuring transparency, ethical awareness, and meaningful human control over data processes.

Un sistema socio-tecnico è un modello di progettazione e gestione organizzativa che studia e integra l'interazione tra le persone (sistema sociale) e i mezzi/tecnologie (sistema tecnico) all'interno di un'azienda o di una comunità.

The Data Centric AI Principles

Designing AI as a sociotechnical system means moving beyond model optimization to support human agency, augment human capabilities, and ensure responsible, fair, and transparent use across different stakeholders.



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The Data Centric AI Principles

Achieving high performance requires collaboration with between AI engineers and domain experts, who provide operational knowledge and context-specific quality standards.
Continuous feedback loops help align AI models with real-world needs, maintain data consistency, and maximize performance, turning annotators and experts into active collaborators rather than temporary service providers.



Methodology in Practice: CRISP-DM

A Machine Learning project requires a structured approach; choosing the best algorithm is only one of the success factors

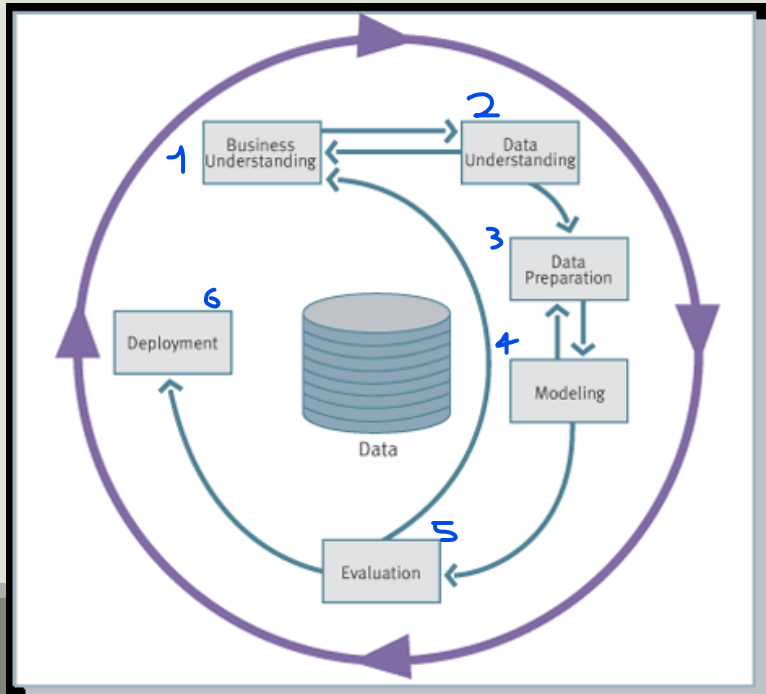
The **CRISP-DM** methodology (proposed first in 1999) is one of the most structured proposals to define the fundamental steps of a Data Mining project

The six stages of the life cycle are not strictly sequential.

Turning back on activities already done it is often necessary

http://www.crisp-dm.org/

Typically, after a few iterations, the understanding phase is closed



With Machine Learning course, we have done only the Modeling part

CRISP-DM steps

1) **Understanding the Application Domain:** understanding project goals from the user's point of view, translate the user's problem into a data mining problem, and define a project plan (Focus on the domain with domain experts)

2) **Understanding the data:** preliminary data collection aimed at identifying quality problems and conducting preliminary analyzes to identify the salient characteristics (analyse data statistically; data give same truth of the experts? Need to know in this phase and talk with experts)

3) After Understanding the data and domain
Data Preparation: includes all the tasks needed to create the final dataset: selecting attributes and records, transforming and cleaning data (like feature engineering)

CRISP-DM steps

The dataset, after preparation, is submitted to the model created and evaluated and if the performance is not good enough we need to go back to previous phase (Domain and data understanding or Data preparation)

4) **Model Creation:** Several data mining techniques are applied to the dataset also with different parameters in order to identify what makes the model more accurate

5) **Evaluation of Model and Results:** The model(s) obtained from the previous phase are analyzed to verify that they are sufficiently precise and robust to respond adequately to the user's objectives

6) **Deployment:** The built-in model and acquired knowledge must be made available to users. This phase can therefore simply lead to the creation of a report or may require implementation of a user-controlled controllable data mining system

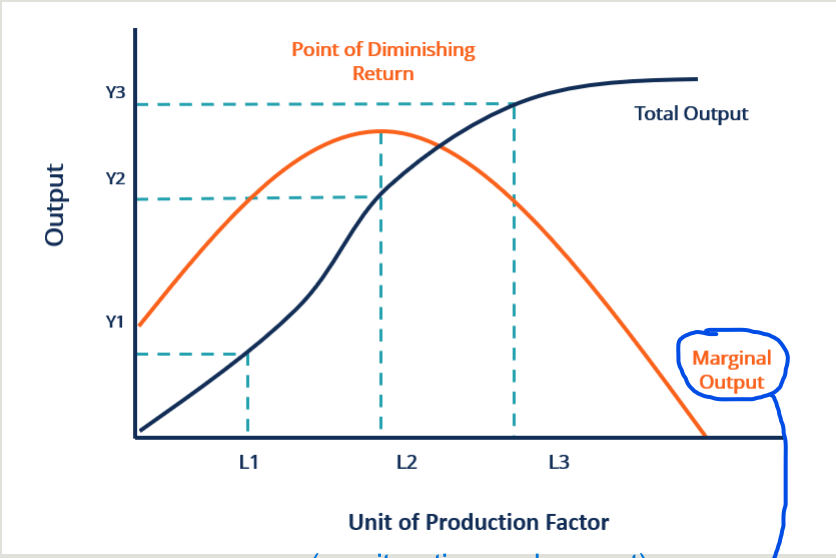
Diminishing Returns in Machine Learning

The diminishing returns principle states that as model performance improves, each additional gain requires disproportionately more resources.

- Analyze the cost-benefit tradeoff and decide how to invest your resources (e.g. time, money)
- Model improvement dominate in the early stage but flatten earlier

The question here is: When should i stop to improve my model?

- In real-projects you must understand when:
  - Performance are good enough — The value of our algo, in most of the cases, is measured as the impact on the domain problem
  - Further improving performance has a positive cost-benefit tradeoff (Need to consider the resources like economic ones)



(e.g. iteration cycle count)

Tasso di miglioramento dallo scorso ciclo iterativo rispetto all'attuale

Diminishing Returns in Machine Learning

In real data science projects:

- There is only one optimal solution but many *good-enough* ones
- Optimal solution may have a worse cost-benefit tradeoff
- You have to try out many alternative paths
- **The hard part is not getting lost**



We can't give you the *perfect recipe*, but we can show you the different options and explain how to combine them!